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Telehealth parent coaching to improve daily living skills for children with ASD

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Children with autism spectrum disorder often display deficits in daily living skills. Behavior analysts can use telehealth, such as videoconferencing technology, to deliver interventions to families of these children. Given the COVID-19 pandemic and the common barriers to accessing behavioral interventions, it is imperative to evaluate the effectiveness and practicality of delivering behavioral interventions via telehealth. This study evaluated the efficacy of a parent-implemented intervention with coaching via telehealth to improve daily living skills. Children ranging in age from 5 to 9 years participated in the study with 1 or 2 of their parents serving as the primary implementer(s). Parents implemented the intervention with fidelity and the intervention yielded increases in independent daily living skill completion for all 4 participants.

Key words: parent-implemented interventions, autism spectrum disorder, adaptive skills, daily living skills, telehealth

Families of children with autism spectrum disorder (ASD) often do not have access to services due to barriers associated with traveling to professionals (Benson et al., 2018; Mello et al., 2016). In addition, there is currently elevated risk associated with the delivery of in-person services during the COVID-19 pandemic (Cox et al., 2020). The use of telehealth (i.e., distance technology) can alleviate many of these and other barriers to accessing services (Boisvert et al., 2010). Some potential benefits of telehealth services include increased access to services for clients living in rural areas, reduced travel costs, increased ease of professional collaboration, and maintained privacy for parents who prefer not to have professionals visit their home (Bice-Urbach & Kratochwill, 2016; Mello et al., 2016; Tomlinson et al., 2018).

Over the last 15 years, research on delivering behavioral interventions using telehealth has increased (Tomlinson et al., 2018). Much of this research is related to increasing social communication skills and decreasing challenging behavior (e.g., Machalicek et al., 2016; Vismara et al., 2012; Wacker et al., 2013). Specific assessment and intervention procedures evaluated in previous research include differential reinforcement, preference assessment, discrete trial teaching, functional analysis, and functional communication training (see Tomlinson et al., 2018). Often a professional coaches a parent using videoconference technology and the parent delivers the intervention to the child (e.g., Machalicek et al., 2016; Suess et al., 2014; Wacker et al., 2013). These studies indicate behavioral interventions delivered via telehealth lead to improvements in child

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outcomes and lower costs of delivering services (Lindgren et al., 2016; Machalicek et al., 2016; Wacker et al., 2013). The use of telehealth to target adaptive skills (e.g., Pellegrino & DiGennaro Reed, 2020) is an emerging area of research.

Broadly, adaptive behavior refers to any skill that promotes independent living (Carter et al., 1996). Individuals with ASD typically have marked deficits in daily living skills such as dressing, housekeeping, and meal preparation; these deficits often persist into adulthood (Liss et al., 2001; Smith et al., 2012). Adaptive skills deficits can interfere with an individual's opportunities to live independently. In addition, independently completing daily living skills (e.g., hand washing) can contribute to obtaining meaningful employment. Few adults with ASD live independently and approximately one third are employed (Howlin et al., 2004). It is likely that deficits in daily living skills contribute to low independence and employment for individuals with ASD. Therefore, delivering telehealth services to address deficits in this area could be an important step forward in evaluating the effectiveness of this mode of service delivery.

Telehealth may be particularly useful for delivering interventions to increase daily living skills. Like the use of telehealth to address other outcomes, telehealth can reduce travel costs, increase flexibility for scheduling sessions, and increase access to services. The increased scheduling flexibility may have added benefits in the case of daily living skills. These skills often occur during times that are difficult for professionals to be in the home. For example, a child may typically brush his teeth in the morning and evening. Some families may not want a professional in their homes during these times. In these cases, a brief telehealth session could be scheduled during the morning or evening routine. In addition, many daily living skills, such as hand washing, are especially important during the COVID-19 pandemic (Alzyood et al., 2020; Centers for Disease Control and Prevention, 2020). For these reasons, it is critically important to evaluate the efficacy of interventions delivered via telehealth in improving daily living skills.

Effective treatment strategies to address daily living skills include chaining procedures (e.g., Jerome et al., 2007; Rayner, 2011), video modeling (e.g., Bidwell & Rehfeldt, 2004), visual schedules (e.g., Krantz et al., 1993; Pierce & Schreibman, 1994), and a combination of these procedures (e.g., video-enhanced activity schedule; Kirkpatrick et al., 2019). Research evaluating effective treatment strategies to address daily living skills within a telehealth model is only beginning to emerge. For example, in a recent study, researchers implemented a total-task chaining procedure to teach two adults with disabilities to engage in daily living and recreational skills (Pellegrino & DiGennaro Reed, 2020). In this study, the researcher delivered the intervention directly to the participants using videoconferencing, which resulted in mastery of targeted skills for both participants. The participants could read written instructions, imitate video models, and engage in reciprocal interactions on videoconference. However, for some participants, it may be necessary to use a parent-mediated approach, in which the professional delivers coaching to the parent over telehealth and the parent implements the intervention with the child. Boutain et al. (2020) demonstrated that parents accurately implemented graduated guidance to teach self-care skills (e.g., hand and face washing) with their children following coaching they received via telehealth. The parent training procedure involved behavioral skills training (i.e., instructions, modeling, rehearsal, and feedback) and required four adults (i.e., two researchers to conduct modeling, the parent, an adult confederate for rehearsal sessions). This training package resulted in high implementation fidelity for all included parents.

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Although the results of these recent studies are encouraging, it is imperative to extend this important line of research, especially given the current COVID-19 pandemic. There is a need to extend the range of procedures that can be taught using a parent-mediated telehealth approach and to evaluate parent training procedures that require fewer resources (i.e., the number of adults present during service delivery). Thus, the purpose of the current study was to evaluate the extent to which a parent-implemented total-task chaining procedure, with coaching delivered via telehealth technology, would lead to an increase in independent completion of daily living skills for children with ASD.

Method

Participants

Four boys, age 5 to 9 years old, diagnosed with ASD, and their parents participated in the study. Participants were recruited as part of a larger project on telehealth coaching with parents of children with ASD. Each family received six or more weeks of parent coaching via telehealth, related to interventions to address three or more individualized goals. To be eligible to participate in the study, three of the child's goals needed to target an increase in daily living skills. Mothers (for two children) or mothers and fathers (for two children) served as the implementer for each session. Table 1 includes additional participant information. None of the children was receiving applied behavior analysis services; all four children received at least one other type of therapy (e.g., speech therapy, physical therapy). None of the parents reported receiving previous training in applied behavior analysis interventions or parent training for their child via telehealth. However, all parents had taken some coursework in interventions, typically related to their career (e.g., teacher, counselor, social worker). The families participated for 8, 6, 5, or 3 months (Ross, Charlie, Henry, and Liam, respectively).

Setting and Materials

The telehealth visits occurred in the participants' homes, in a room selected by the parent. Therapists provided coaching from a room on a university campus or from their own homes during the COVID-19 pandemic. Materials related to the daily living skill (e.g., soap for handwashing) were present during every session. In addition, highly preferred items, identified via a free operant preference assessment (Roane et al., 1998), and/or food were available during intervention sessions.

Technology

During the study, the therapists used laptop computers with a camera, speakers, and a built-in microphone (or headphones, based on the therapist's preference) to provide coaching via videoconference. The videoconference application, VSee®, was used to conduct all sessions; this program uses encryption and point-to-point transmission (VSee, 2020). These features protect participant confidentiality and can support compliance with HIPAA (VSee, 2020). VSee also requires less bandwidth than other videoconference software (VSee, 2013).

At the beginning of the study, we asked all participating families if they would like to receive videoconference technology for use during the study. Families chose this option if they did not have the required technology or access to internet, or if they believed their technology was not appropriate for any reason (e.g., technology was out of date, the technology could not be easily moved from room to room). Liam's and Henry's families elected to receive videoconference technology, while the remaining two families used their own technology. For Liam and Henry's families, we sent an iPad® tablet, a charger and stand for the tablet, and a Bluetooth® headset to the home. Parents

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Table 1

Child Participant Information

Child (Parent Implementer)	Age	Diagnosis	Targeted Adaptive Skills (Number of Steps)
Ross (mother)	9-1	ASD, intellectual disability, hypotonia, speech impairment, infantile seizures	• Wiping off the counter (8) • Setting the table (8) • Unloading the dishwasher (21)
Charlie (mother)	6-5	ASD, 22q11.2 deletion syndrome	• Cleaning up toys (10) • Taking off shirt (6) • Washing hands (9)
Henry (mother and father)	5-11	ASD	• Cleaning up toys (10) • Washing hands (9) • Putting on shoes (8)
Liam (mother and father)	6-4	ASD, speech delay	• Cleaning up toys (9) • Using a fork (6) • Washing hands (9)

Note. Age is presented as years-months.

had the option to use the headset during visits, but neither family did. The iPad had a case and screen protector. We downloaded the VSee application onto the iPad prior to sending it. In addition, cellular service was purchased for the iPads, which could be used for the visits, so that families did not need to have wireless internet to participate in the study. The remaining two families used technology they already had available in the home. Both families used iPads with a stand and typically connected to the visits via wireless internet. The families selected the technology used; the only requirements were that the technology had video and audio (microphone and speaker) capabilities. We also recommended that the video could capture the area where the sessions occurred without the parent holding the technology, so that parent movement was not restricted throughout the session.

Prior to the start of the program, the therapist helped families set up the technology. The technology was mailed directly to the family's homes along with instructions for setting up and returning the technology. Families who used their own technology received an invitation to register for and download VSee prior to the first session. Parents received support for using the technology (e.g., providing instructions on the phone as they set up their technology) throughout the study and parents typically needed the most support during the first visit.

During visits, the therapist asked parents to position the technology in a location that allowed the therapist to see the participant and all relevant materials for each session. The therapist ensured that all necessary materials were within the frame prior to the parent beginning the session. For some sessions, parents needed to follow the child with the camera (e.g., during goals that involved moving from room to room, if the child walked away). The video quality very rarely interfered with coaching or data collection. If the therapist could not adequately view the behavior or the parent could not hear the therapist (e.g., due to pauses in the video or audio, inadequate video quality), the therapist instructed the parent to conduct the session again. The therapist repeated instructions or feedback as necessary if the video or audio quality interfered with coaching.

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Data Collection

Dependent Measure

The therapist and supervising Board Certified Behavior Analyst® (BCBA®) created the task analysis for each skill and altered the task analysis as needed based on the parent's feedback. The therapist recorded the child's behavior as correct or incorrect for each step in the task analysis and then calculated the percentage of steps completed correctly. Correct completion of a step was defined as independently completing the step within 5 s following the instruction (for the first step) or the completion of the previous step (for all remaining steps) for each session. The child needed to complete the steps in the correct order for the step to count as correct. All other responses, including prompted responses, waiting more than 5 s to initiate the step, and engaging in a different behavior (i.e., not the appropriate step in the task analysis) were counted as incorrect. There were 6 to 21 steps per task analysis (see Supplemental Materials for task analyses). We collected data on the dependent measure during baseline and intervention sessions; we did not collect data on child behavior during the teaching trials due to the use of immediate physical prompting in these sessions (i.e., no opportunity for the child to engage in an unprompted response).

Interobserver Agreement

A second, independent data collector recorded data on daily living skills for more than 80% of the sessions in each phase of the study and for each participant. These data were collected during the session or via video recording. Both data collectors were in the same room or in their own homes (during the COVID-19 pandemic). During the COVID-19 pandemic, the data collectors joined the session independently from their own computers. We scored the step as an agreement if both data collectors recorded the same step as correct or incorrect. We then calculated the percentage of steps with agreement for each session. The average percentage agreement was 98% (range, 67%–100%), 99% (range, 67%–100%), 99% (range, 71%–100%), and 98% (range, 73%–100%) for Henry, Charlie, Ross, and Liam, respectively.

Parent Implementation Fidelity

The therapist or observer recorded the percentage of steps implemented correctly by the parent during at least 90% of sessions for each child. We developed a task analysis of the procedures for each phase of the study. For baseline, the steps consisted of delivering the instruction (e.g., "wash your hands"), not prompting the child to complete the steps, and providing a neutral statement (e.g., "you're all done") or no statement at the completion of the session. For the teaching trials and intervention, the steps consisted of providing the instruction, correctly using the prompting procedure, and providing praise and the reinforcer following completion of the skill. A parent had to implement each step correctly every opportunity during the session for the step to be scored as correct (e.g., the observer scored the prompting fidelity step as incorrect if the parent implemented the prompting procedure incorrectly for one or more steps of the daily living skill).

For baseline, parent implementation fidelity was 94% (range, 67%–100%), 81% (range, 0%–100%), 100%, and 100% for Henry, Charlie, Ross, and Liam, respectively. For teaching trials, parent implementation fidelity was 96% (range, 67%–100%), 86% (range, 67%–100%), 100%, and 100% for Henry, Charlie, Ross, and Liam, respectively. For intervention sessions, parent implementation fidelity was 89% (range, 33%–100%), 87% (range, 33%–100%), 97% (range, 67%–100%), and 96% (range, 67%–100%) for Henry, Charlie, Ross, and Liam, respectively. Table 2 summarizes parent implementation fidelity by step. The most frequently missed step during intervention sessions was the prompting step.

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Table 2

Parent Implementation Fidelity for the Intervention Sessions by Step

Child	Instruction	Prompting	Reinforcement
Ross	100%	93.20%	97.10%
Charlie	97.30%	72.60%	94.52%
Henry	97.90%	81.00%	87.23%
Liam	100%	88.00%	100%

Coaching Fidelity

To evaluate the extent to which the therapist correctly implemented the coaching procedures, we calculated the percentage of coaching steps implemented correctly during 84% of sessions. Coaching fidelity consisted of steps at the beginning of the visit, during each session, and at the end of the visit. At the beginning of the visit, the therapist asked how the parent and child were doing, asked about conducting the sessions between visits (if applicable), and gave an overview of the plan for the visit. During each session, the steps included providing instructions, providing prompting, and providing praise or feedback (depending on the condition). At the end of the visit, the therapist asked the parent if they had any questions, reminded the parent about the next visit, and recommended practicing the sessions between visits (if applicable). Average coaching implementation fidelity was 99% of steps completed accurately across sessions (range, 60%–100%).

Therapist Training

The therapists who provided the majority of the parent coaching were graduate research assistants enrolled in master's level coursework for the BCBA® certification. Prior to their enrollment in the master's program, they had not been employed to deliver behavioral interventions to children with developmental disabilities; they had no experience with telehealth interventions or training parents. A BCBA or BCBA-Doctoral® (BCBA-D®) with experience implementing telehealth behavioral interventions provided training and supervision using a behavioral skills training (BST) model (Miltenberger, 2016). Prior to the initial training, therapists completed online modules describing evidence-based practices for children with ASD. The initial training (written and verbal instructions, modeling, and rehearsal with the instructor) required approximately 2 weeks (approximately 20 hr per week). This training included teaching the correct implementation of the session procedures and telehealth parent coaching. Through rehearsal with the instructor, each therapist demonstrated correct implementation of all session procedures. Immediately following training, the therapists received supervision during all or most of their sessions with families. The therapists received supervision from a BCBA or a BCBA-D (called "BCBA" throughout) during the study.

Experimental Design

We used a concurrent multiple-baseline design across skills (Kennedy, 2005) to evaluate the efficacy of the intervention. Each tier consisted of baseline, teaching trials, and intervention. The therapists conducted visits 1 to 3 days per week, for approximately 1 hr each. We scheduled visits between 8 AM and 8 PM, Monday through Saturday. The therapist scheduled visit times based on the family's preference and the therapist's availability. Up to four sessions per skill were conducted per visit. A session consisted of one trial for the skill. The therapist or an observer collected data during each session or via the video recording.

Procedures

Parent Training

Parent training consisted of instructions, prompting, and feedback on session performance. We developed the parent training procedures based on the BST model (Miles &

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Wilder, 2009; Ward-Horner & Sturmey, 2012); we prioritized procedures that required little time and did not require additional adults to be present.

With the introduction of any new procedures (i.e., preference assessment, baseline, teaching trials, intervention), the therapist provided written and vocal instructions that included a rationale for the procedure and the steps required to implement the session. The therapist shared the instructions via screen sharing or file sharing in the VSee application. All written procedures were free of behavior analytic jargon and written as simply as possible (i.e., numbered step-by-step format, short statements). The therapist described the procedures based on the instructions and answered all parent questions. On subsequent visits, the therapist quickly reviewed the instructions before asking the parent to conduct sessions. This review consisted of vocal instructions; if the therapist preferred to include written instructions, the therapist also provided the written instructions during the review.

Following the instructions, the parent then implemented the relevant procedures with their child (preference assessment, baseline, teaching trial, or intervention). Throughout these sessions, the therapist vocally described how to correctly implement the procedures. The therapist provided a specific verbal prompt if the parent began a step incorrectly or did not initiate a step. Following the preference assessment, baseline, and teaching trials, the therapist provided praise (e.g., "that looked good," "you did great"). After each session during intervention, the therapist provided performance feedback, including praise for steps completed correctly and a description of how to correctly implement steps they completed incorrectly. Therapists provided performance feedback following sessions during the intervention condition, and not during other phases, because we expected parents to continue implementing intervention procedures after the study ended. Once a parent reached 80% implementation fidelity in the intervention phase, the therapist encouraged them to practice the skills between sessions. The therapist asked parents to provide anecdotal reports on progress between sessions. The therapist did not collect data on parents' implementation of the between-visit sessions; similarly, the therapist did not ask parents to collect or report child data from these sessions.

For Henry's and Liam's families, both parents implemented sessions. Typically, one parent conducted all sessions during the visit. The family decided which parent implemented the sessions based on parent preference and availability. For example, Liam's father typically worked during the weekday sessions, so the mother often implemented weekday sessions and the father typically implemented sessions on the weekends.

Goal Development and Preference Assessment

Prior to the start of baseline, the therapist and BCBA conducted a parent interview, observation, and preference assessment. The parent interview consisted of a researcher-developed interview and the Vineland Adaptive Behavior Scales Third Edition (Sparrow et al., 2016). In the subsequent visit, the parent and child participated in a 10-min observation during which the therapist instructed the parent to play with the child. The observation provided information regarding the child's communication, play skills, and problem behavior, which informed the list of goals developed by the therapist. Following the interview and observation, the therapist developed a list of goals. During the following visit, the therapist and parent discussed the list of goals and the parent selected three or more goals based on their priorities; the parent could select goals from the list, change goals on the list, and add new goals. The use of telehealth influenced the types of skills selected. For example, we reminded the parent that the child would be on video during the sessions, so we recommended avoiding certain

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skills (e.g., showering). In addition, we only targeted skills conducted in the home due to potential barriers associated with conducting sessions in the community (e.g., internet connection, presence of other individuals).

The parent also conducted a 5-min free operant preference assessment (Roane et al., 1998) to identify highly preferred items. The preference assessment typically included five items with which the child liked to play based on parent report. The item(s) with the highest percentage of intervals with engagement were selected for use as potential reinforcers during the sessions. The parent conducted the preference assessment prior to the start of baseline and during the study as needed, if observation indicated the item was no longer highly preferred.

Baseline

During baseline sessions, the parent presented the instruction (e.g., "wash your hands"). The parent did not provide any assistance in completing the skill. If the child did not initiate the next step or any other step in the task analysis within 30 s, the session was terminated. At the end of the session, the parent said a neutral statement (e.g., "you're all done") or did not make a statement.

Teaching Trials

The parent conducted teaching trials before and during the intervention phase. Teaching trials included immediate prompting (rather than the constant time delay procedure used during intervention; see below). For some skills, the parent conducted additional teaching trials during the intervention phase if visual analysis indicated no clear increasing trend in the data. In each case (prior to or during intervention), the parent conducted three teaching trials. For teaching trials, the parent presented the instruction at the beginning of the trial (e.g., "wash your hands"), then provided immediate hand-over-hand physical prompts for each step of the daily living skill. After the last step, the parent provided praise and access to a highly preferred item for at least 1 min and/or food (e.g., for the eating with a fork goal).

Intervention

For intervention sessions, the parent presented the instruction (e.g., "wash your hands"). The parent used a constant time delay procedure with a physical prompt for all steps in the task analysis. That is, the parent waited 5 s following the instruction or completion of the previous step for the child to complete the next step in the task analysis. If the child did not complete the step, the parent immediately provided a hand-over-hand physical prompt for that step. Therefore, during intervention, the child had the opportunity to complete each step independently. After the last step, the parent provided praise and access to a highly preferred item for at least 1 min and/or food (e.g., for the eating with a fork goal). The intervention procedures, including the prompting procedure, were the same for all participants. The mastery criterion was 100% of steps completed correctly for the skill for two or more consecutive sessions.

Results

Figures 1, 2, 3, and 4 include the percentage of steps completed correctly for Ross, Charlie, Henry, and Liam, respectively. The intervention increased independent completion of two to three of the three targeted daily living skills for each participant. Overall, participants achieved mastery criterion for seven of the 12 targeted skills. For Ross's treatment evaluation, the intervention produced increases in the dependent variable for all three skills; however, he did not meet the mastery criterion for any of the three skills. For Charlie's treatment evaluation, the intervention produced increases in the dependent variable for two of the three skills (Cleaning up Toys and Taking off Shirt);

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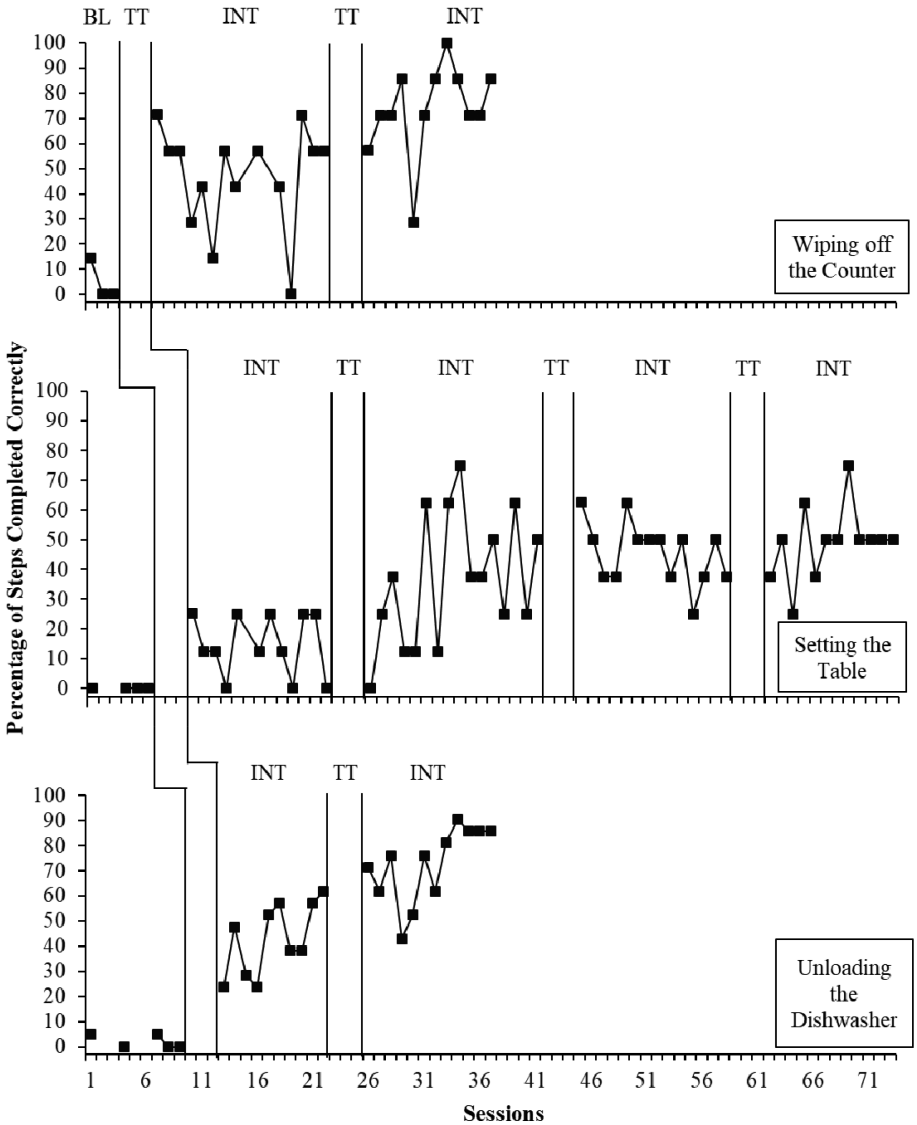
Figure 1
Percentage of Steps Completed Correctly on Each Daily Living Skill for Ross



Note. BL = Baseline, Teaching trials = TT, intervention = INT.

he had a high percentage for Washing Hands during baseline. Charlie met the mastery criterion for all three skills. For Henry's treatment evaluation, the intervention produced increases in the dependent variable for two of the three skills (Cleaning up Toys and Washing Hands) and mastery for one of the three skills (Cleaning up Toys). We offered to continue conducting sessions with Henry's family because he had not met the mastery criterion for two skills, but Henry's family declined due to personal difficulty associated with the COVID-

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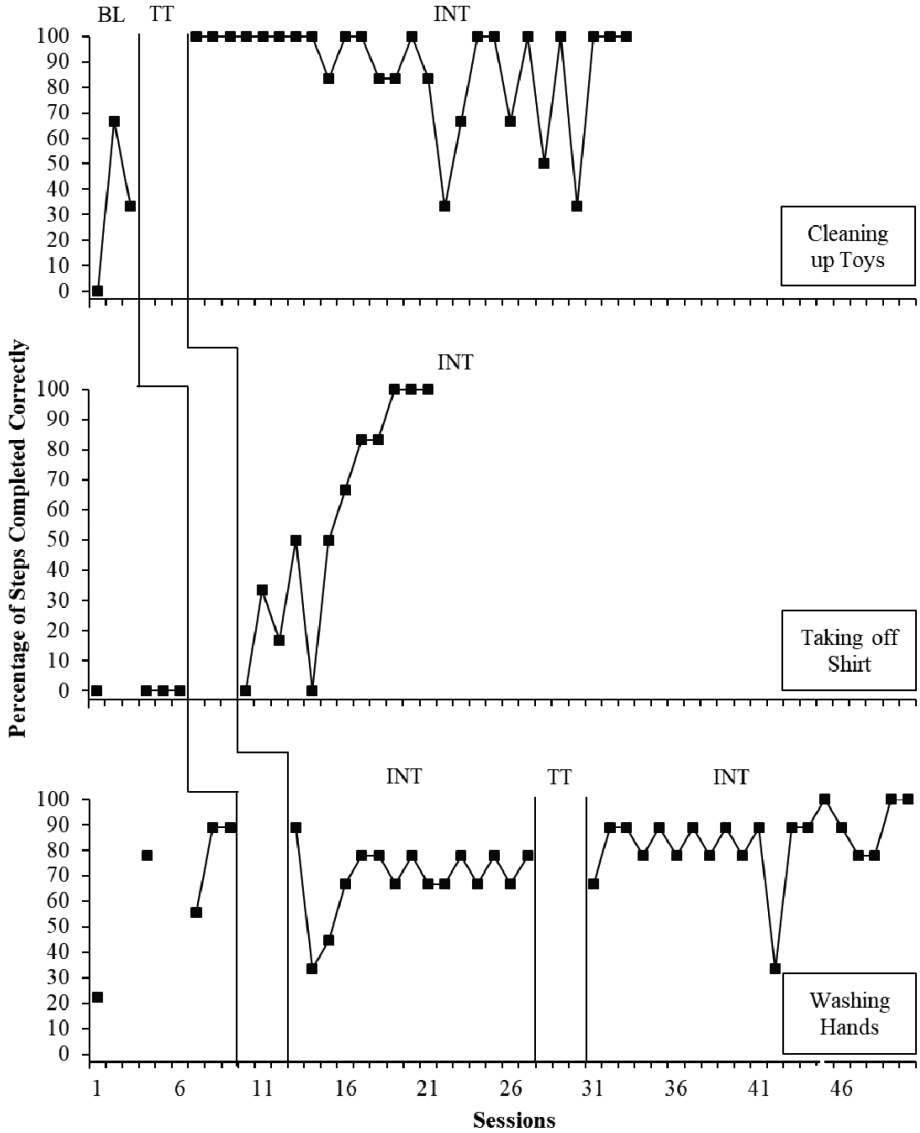
Figure 2

Percentage of Steps Completed Correctly on Each Daily Living Skill for Charlie



Note. BL = Baseline, Teaching trials = TT, intervention = INT.

19 pandemic. For Liam's treatment evaluation, the intervention produced increases in the dependent variable for three of the three skills; he met the mastery criterion for all three skills. Across all participants, the average percentage of steps completed correctly during baseline was 19% (range, 0%–89%) and during intervention was 66% (range, 0%–100%). Together, the data across all participants indicated that the parent-implemented intervention yielded increases in daily living skills.

Two of the participants (Charlie and Henry) engaged in high rates of correct responding

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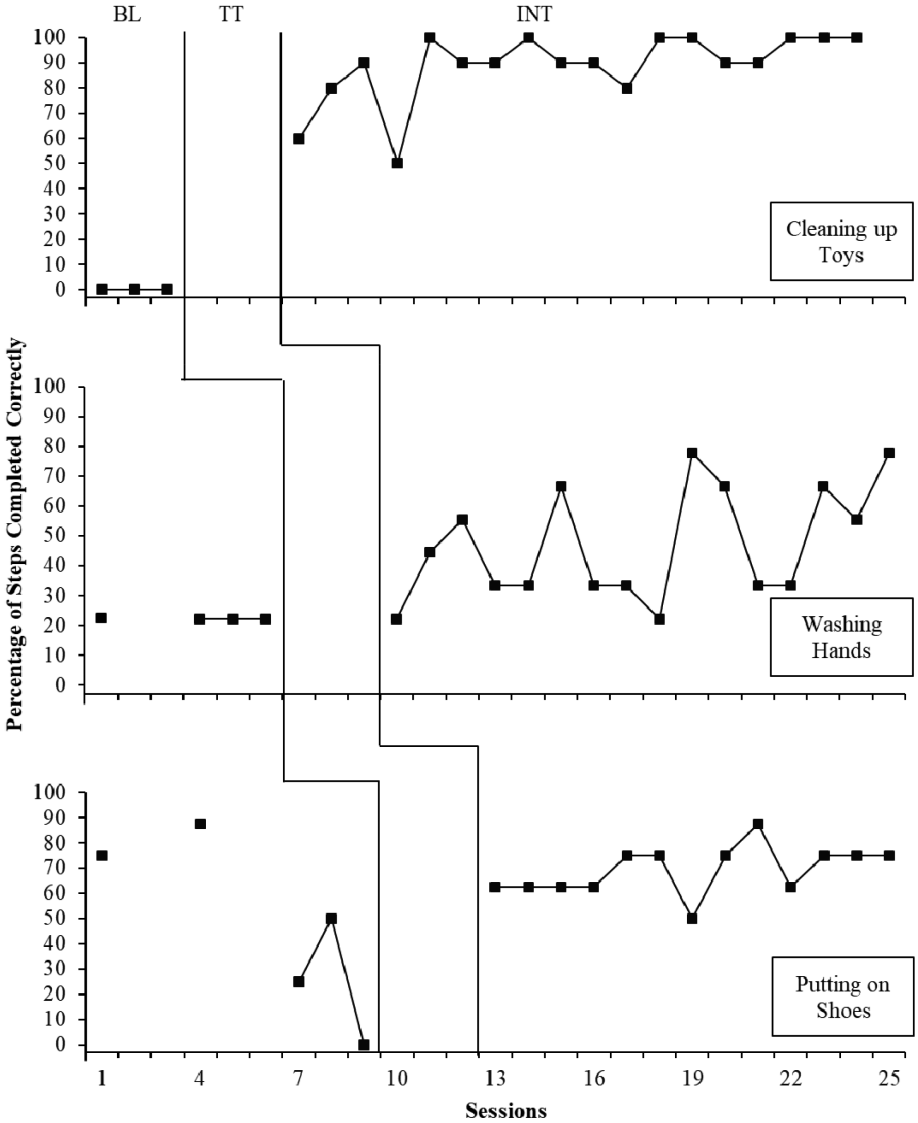
Figure 3

Percentage of Steps Completed Correctly on Each Daily Living Skill for Henry



Note. BL = Baseline, Teaching trials = TT, intervention = INT.

during baseline for one skill, making it difficult to evaluate the extent to which correct responding improved during intervention. For the Washing Hands skill, Charlie's baseline data were variable but high on average (*M* = 67%; range, 22%–89%). The percentage of steps completed correctly remained variable during the first intervention phase (*M* = 69%; range, 33%–89%). Following the second set of teaching trials, Charlie achieved 100% accuracy for two consecutive sessions. For Putting on Shoes, Henry's data were relatively high and variable during baseline, with

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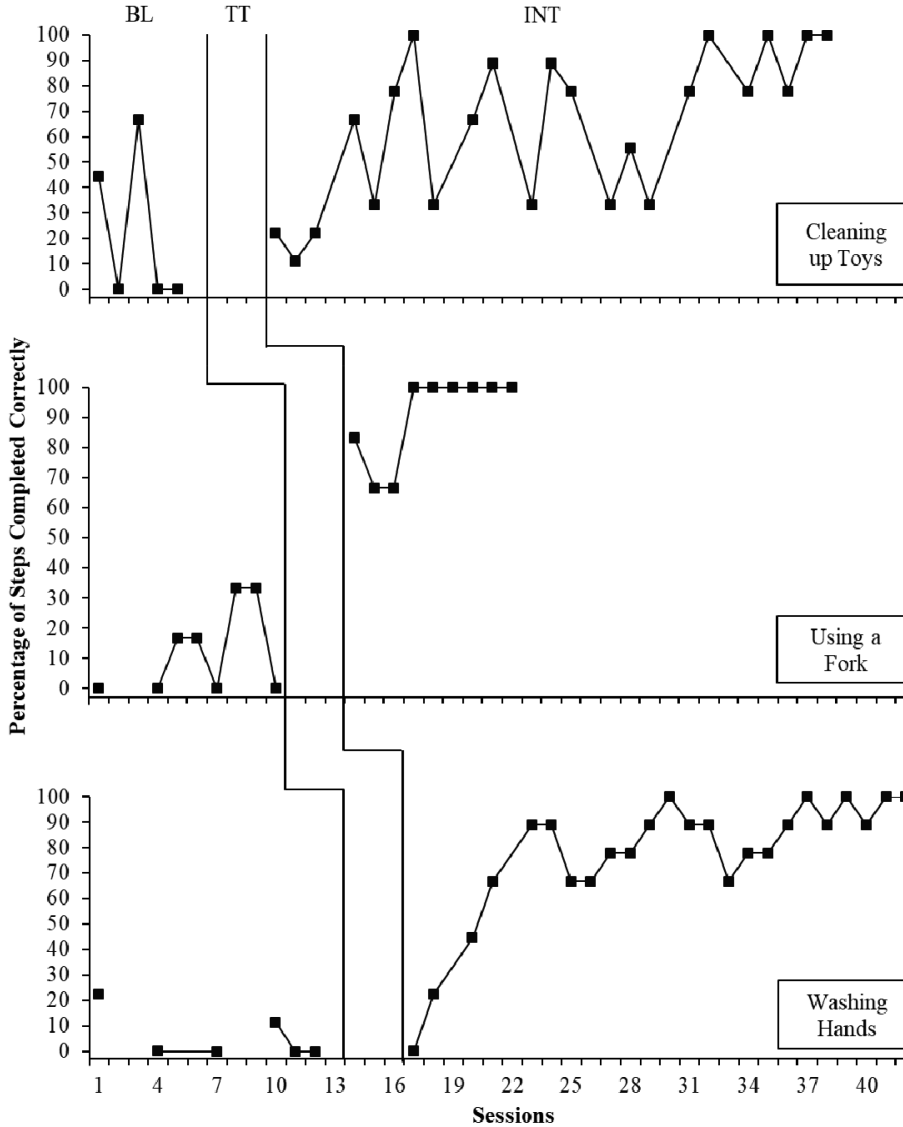
Figure 4

Percentage of Steps Completed Correctly on Each Daily Living Skill for Liam



Note. BL = Baseline, Teaching trials = TT, intervention = INT.

a decreasing trend over the final three baseline data points (M = 48%; range, 0%–88%). During the first two baseline sessions, the parents positioned the shoes in front of the child, with the corresponding shoe next to each foot, which was not part of the procedure for baseline. For the remaining sessions (baseline, teaching trials, and intervention), the therapist instructed the parent not to arrange the shoes in front of the child. The intervention was associated with greater stability in the data, but there was no clear increase in level (M = 69%; range, 50%–88%).

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The therapist conducted additional teaching trials if the data indicated no trend or a decreasing trend. The purpose of these trials was to promote skill acquisition by conducting sessions with additional support (i.e., immediate full physical prompting). However, the efficacy of this strategy remains unclear, due to variability in the data and lack of immediacy of effect following the additional teaching trials. For Ross, the parent conducted additional teaching trials for all three skills. In the initial intervention phase, the level increased for all three skills compared to baseline. However, Ross did not meet the mastery criterion and the data indicated little or no increasing trend for each of the skills in the first intervention phase. We then implemented a second set of teaching trials across all three skills. Subsequently, the level increased in the second intervention phase for all three skills, but Ross did not meet the mastery criterion for any of the skills. Due to the lower level and stable trend in the Setting the Table skill, the parent conducted additional teaching trials for this skill. However, there was no clear increase in level following the third and fourth set of teaching trials. Charlie did not meet the mastery criterion and there was no increasing trend for the Washing Hands skill. Therefore, we implemented a second set of teaching trials. Following those teaching trials, Charlie achieved 100% correct for two consecutive data points.

Discussion

The purpose of the present study was to evaluate the efficacy of a parent-implemented intervention with coaching delivered via telehealth in improving daily living skills. The study extends recent work in the area of telehealth adaptive skills interventions (Boutain et al., 2020; Pellegrino & DiGennaro Reed, 2020). For this study, four children with ASD and their parents participated. The intervention produced increases in the percentage of steps completed correctly for 10 of the 12 skills addressed. These results indicate that parent-implemented interventions, with coaching via telehealth, can yield increases in daily living skills. Furthermore, parent implementation fidelity data indicate that parents accurately implemented the intervention.

The findings of this study are important given the utility of delivering coaching via telehealth to address daily living skills. First, it is often difficult for therapists to travel to homes, which are the natural setting to teach many daily living skills. In cases in which therapists can travel to the child's home, it can be difficult to arrange session times that correspond with when the specific daily living skill will be performed. The use of coaching via telehealth can ameliorate many of the common barriers to implementing interventions to address daily living skills. Furthermore, a portion of this study took place during the COVID-19 pandemic. This pandemic highlighted the importance of developing and evaluating treatment protocols that do not require face-to-face contact with professionals. This study contributes to the growing literature on the use of telehealth technology to improve daily living skills.

Based on the implementation fidelity and child data, parents implemented the procedures accurately and their implementation of the intervention yielded increases in daily living skills. All parents implemented the baseline, teaching trials, and intervention procedures with high fidelity (average higher than 80%). These results are particularly promising given the few components included in parent training (instructions, prompting, and feedback) and that the therapist never conducted a face-to-face training with the parents. Although we only collected parent implementation fidelity data during coached sessions, previous research indicates parents generalize implementation of interventions to independently conducted sessions (Suess et al., 2014). The results of this study, along with previous research, provide promising

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support for the efficacy of parent-implemented interventions with coaching via telehealth.

We designed the procedures so that families who did not have access to videoconferencing technology or internet could still receive telehealth coaching and participate in the study. Two families chose to receive technology. The procedures included sending a tablet with cellular service to the family, which had associated costs for the research team, including the cost of sending and returning the technology, the cost of the technology, and the cost of cellular service. Additionally, there was a provider subscription fee for VSee; there was no fee for the families. There are also potential risks associated with these procedures, such as loss, damage, or failure to return the technology. We did not experience any of these with the participating families. Practitioners and researchers may want to consider these risks as they develop telehealth procedures. In addition, providing the technology may not be feasible in the context of long-term service provision, in which the families would use the technology and cellular service for multiple years. Previous research indicates that telehealth services to treat challenging behavior cost less than in-person services (Lindgren et al., 2016). However, there is a need for an analysis of costs, risks, and benefits of providing the needed technology in the context of interventions to increase daily living skills and comprehensive behavior analytic services.

It is important to interpret the results of this study with caution and discuss directions for future research, given the limitations of the study. First, for Charlie and Henry, high baseline data with one skill made it difficult to evaluate intervention efficacy across the multiple-baseline design. However, the study included four participants, each with a multiple-baseline design across skills, allowing for several replications of the effect of the intervention. Second, it would have been beneficial to collect generalization, maintenance, and social validity data. Third, for Henry and Liam, both parents conducted sessions; the parent who implemented each session varied based on parent availability and preference. We used this procedure to improve feasibility for the families who wanted both parents involved. It is possible that differences in implementation of the intervention or in the histories of reinforcement caused variability in the data. We did not evaluate the extent to which the implementer affected child data because we did not systematically choose the implementer for each session. Fourth, we selected the same prompting procedure for all participants (i.e., time delay followed by a physical prompt). A physical prompt may not be appropriate for some individuals and the parent or child may prefer a model or visual prompt rather than a physical prompt. Practitioners and researchers should develop and evaluate methods for individualizing prompt procedures within chaining interventions, based on child skill and parent and child preference. Finally, because we required the child to complete the steps in order to count as correct, we may have underestimated correct responding during baseline. We collected data in the same manner throughout the study; therefore, this requirement should have resulted in similar underestimation of correct responding throughout the study. Furthermore, during baseline, none of the participants completed steps out of order and still completed the entire skill, suggesting that the lack of correct completion of the task during baseline was not due to the order requirement.

We encountered some barriers related to the delivery of services via telehealth. The therapists reported that for some participants, parts of the house had poor internet connection. In addition, the video or audio feed would occasionally interfere with the session. In these cases, the therapists typically asked the parent to move to a different room, tried calling the parent again, and/or tried changing the method of connection (e.g., wireless internet instead of cellular service). The therapist restarted some of the sessions due to the technology interfering with data collection. As noted previously, Henry's family did

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not continue participation due to the changes in family life associated with the COVID-19 pandemic. As is the case with scheduling in-person visits, families may have difficulty finding time to schedule telehealth sessions. This may be especially true for times when the family routine is altered, as takes place with the COVID-19 pandemic, long-term health issues, the end of school, or a return to school. Parents also considered siblings' schedules when selecting visit times. For families with multiple children, families frequently selected times in which two parents were home or tried to identify times in which the siblings were likely to be occupied (e.g., the siblings were at school or napping). For example, Liam's parents often arranged other activities for the siblings during the visits to minimize distractions for Liam and the parents. Ross's family scheduled visits for after school, when all of the siblings were home. His older siblings did not need constant supervision; thus, the parent could supervise Ross and his younger sibling at the same time during visits. Practitioners may experience similar barriers in the delivery of telehealth services. Future research should further evaluate therapist's and family's experiences during telehealth visits. Despite encountering barriers related to the delivering services via telehealth, the findings of this study are promising and suggest telehealth interventions can yield improvements in outcomes for children with ASD.

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